

PERIOD-3 ELEMENTS

{Lesson 1}

COURSE OUTLINE;

(a) List of Period 3 elements

- Atomic number
- Electronic configuration

(b) Bonding & structure

(c) Periodic Properties & their periodicity

(d) Other physical properties e.g. physical states, density, etc

(e) Elements

(f) oxides

(g) chlorides

React $\rightarrow$ H_2O $\rightarrow$ dilute Acids

$\rightarrow$ Alkalies

$\rightarrow$ concentrated

Involves Graph work

Structure, bonding & melting point

(h) Mg^{2+} vs Al^{3+}

(i) Diagonal Relationships

(a). SCIENCE SCHOLARS ACADEMY

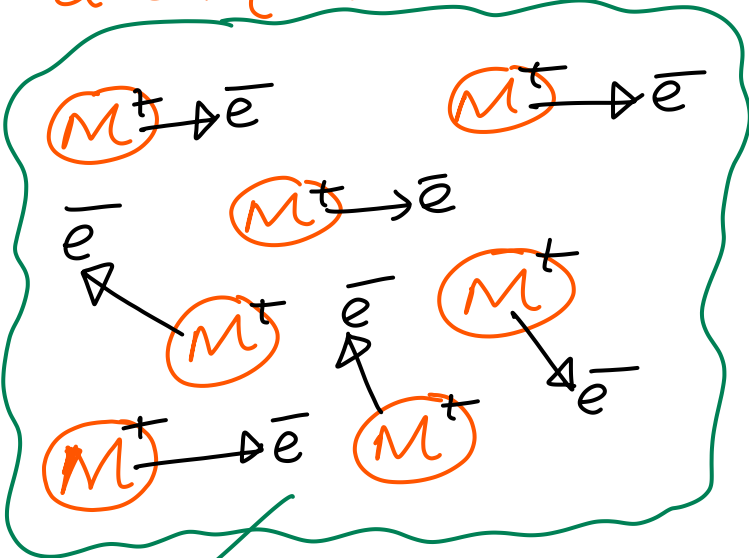
List of Period 3 elements: ✓✓
↳ Refer to a certain book!

(b). Bonding & structure:

Metals (M) (BROADLY)

Metal ^{Metallic} Bond Metal

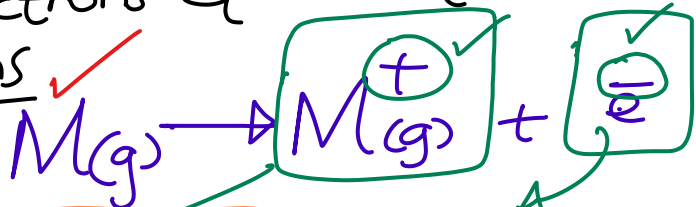
what if we want a complex structure?



{ GIANT ✓
METALLIC ✓
STRUCTURE! }

OR

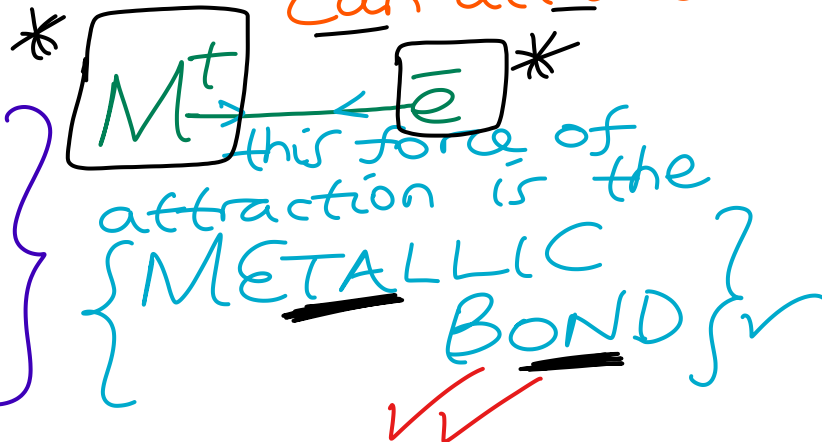
Non-Metals (Nm) ✓
Remember that metals usually lose ✓
electrons & form positive
ions ✓



Positive ✓
ion of
the metal.

electron
cloud
(Negative) ✓

Can attract

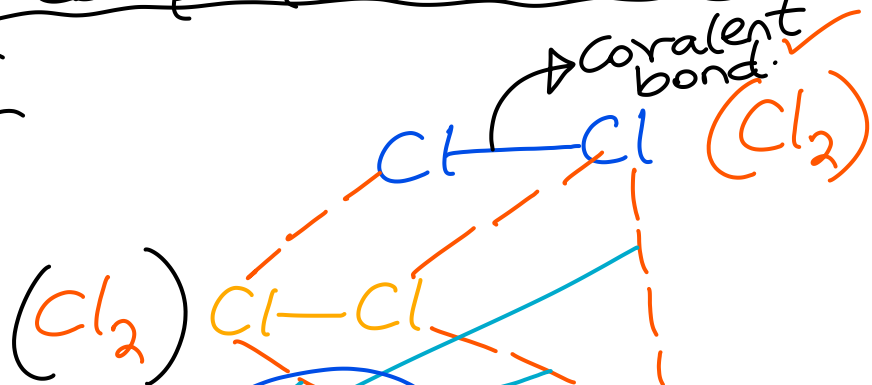
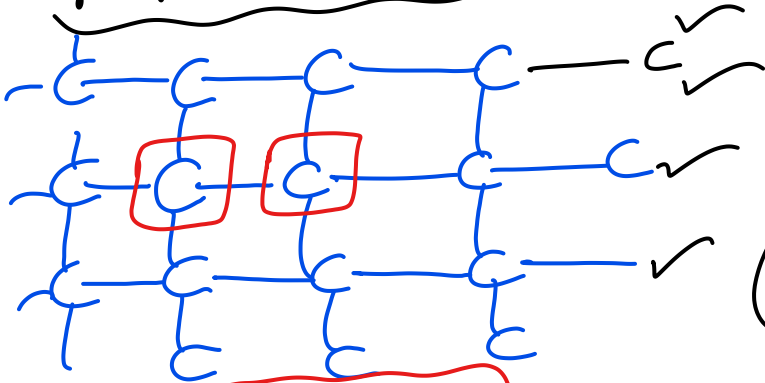


SCIENCE SCHOLARS ACADEMY

NON-METALS



What if we want complex structures?



An endless network of Covalent bonds!

van der Waals forces

Scientist: Johannes Diderick van der Waals

Strong network

Weak network

GIANT

Covalent structure

SIMPLE

Covalent structure

SCIENCE SCHOLARS ACADEMY

(C). Concerning Periodic Properties & Periodicity → Refer to the 2 Lessons Under "periodic table" in the topic of ④ PERIODICITY ① under S.S Term 2 work! ✓

After Going Through Them

→ we shall handle Melting points & Graphical Periodicity!

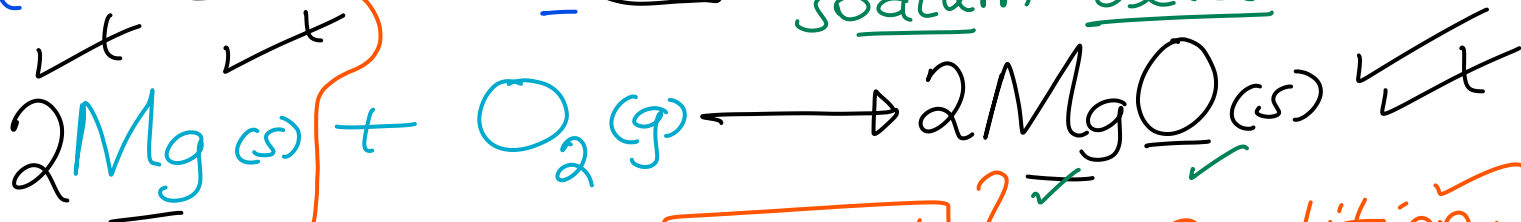
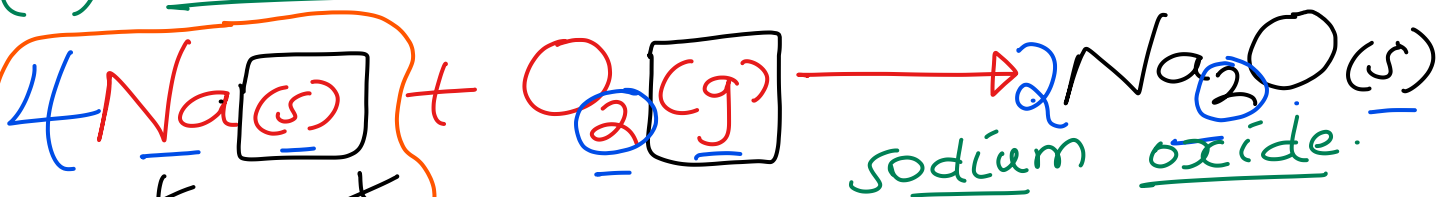
Research: Physical states ✓ & density ✓ of Period 3 elements!

CHEMICAL PROPERTIES
REACTIONS OF
PERIOD-3 ELEMENTS! ✓



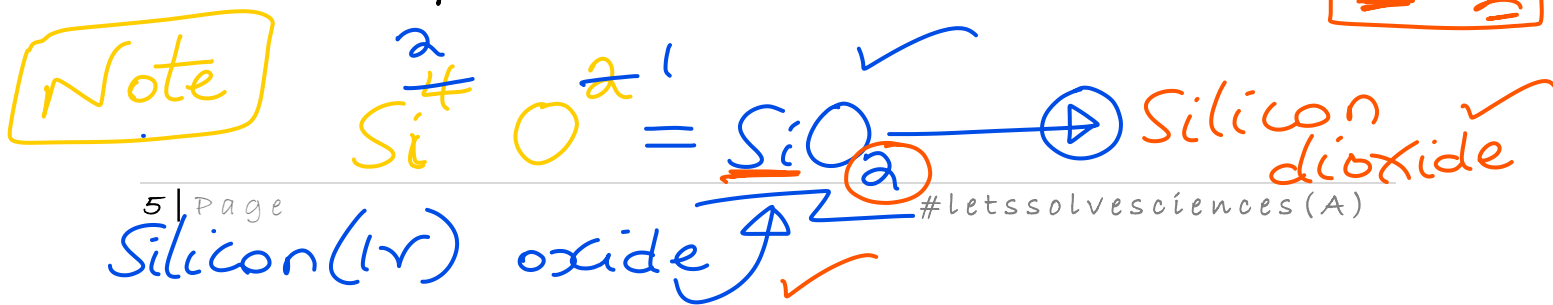
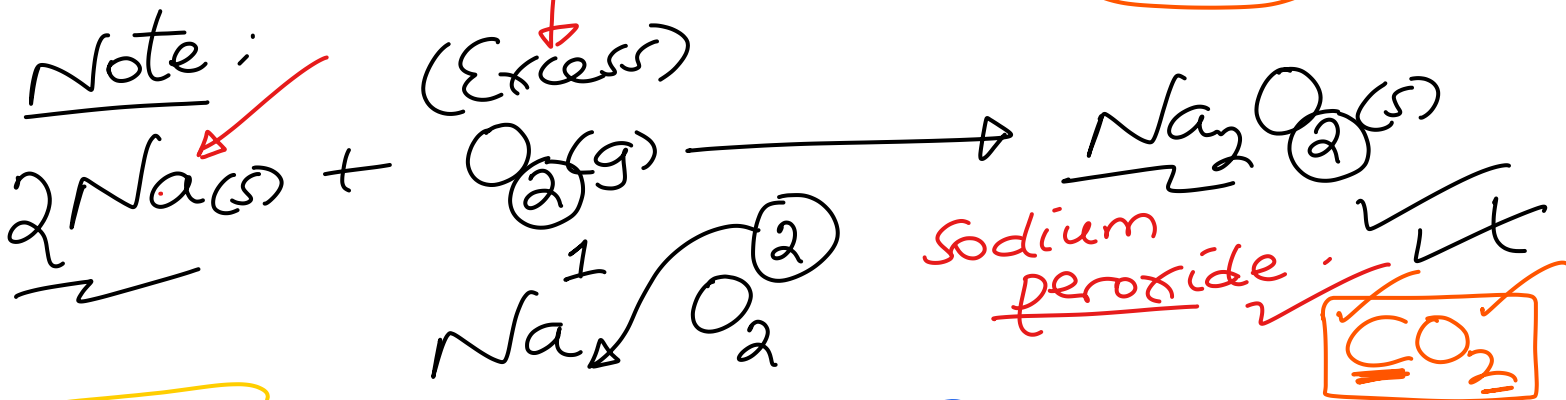
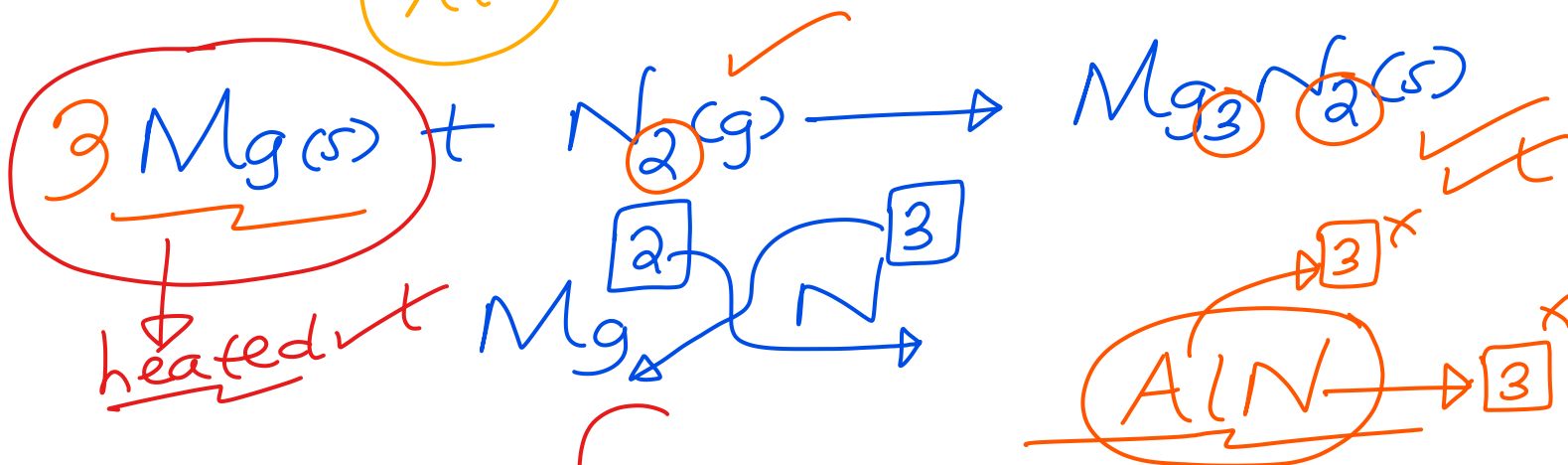
SCIENCE SCHOLARS ACADEMY

(1). Reactions with Air $\rightarrow O_2 \text{ \& \& } N_2$



Metals (Must be heated) $\rightarrow$ Condition.

(* Note: Mg, Al $\rightarrow$ Best friends with nitrogen.



SCIENCE SCHOLARS ACADEMY

Recall → Phosphorus is tetra-atomic ✓

At A-level → P ✗
 → P_4 ✓ for phosphorus.

→ It's a powder ✓

Note: $[15]$ → 2, 8, 5

$P_4(s)$ → Solid

→ tetra-atomic

uses

valency
 $\overset{+3}{P}$ ✓

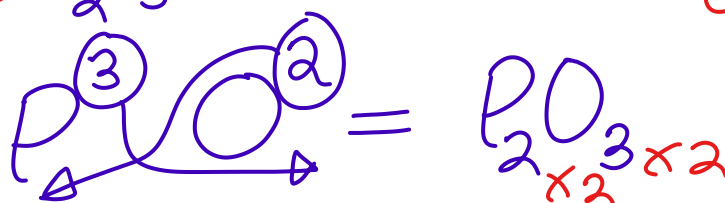
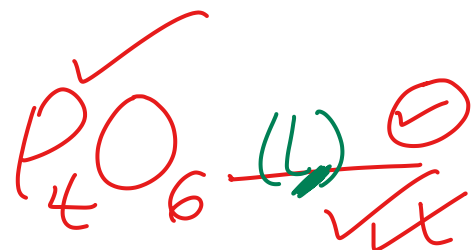
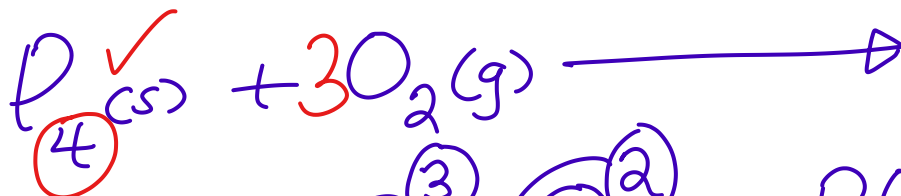
Reacting with

LIMITED O_2 (Cl_2) etc

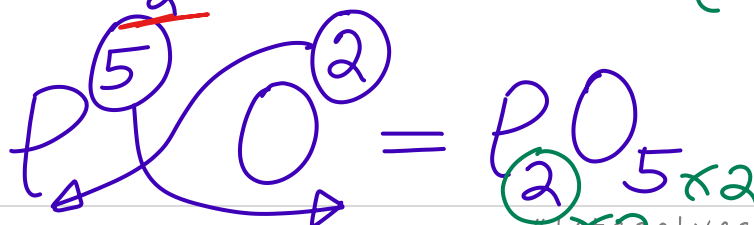
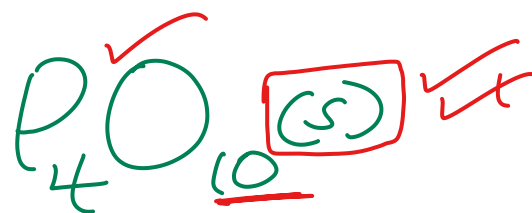
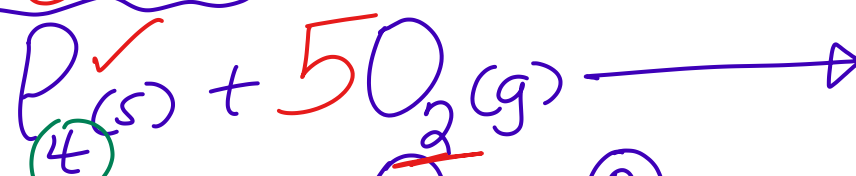
valency of 5 ✓

EXCESS

Limited O_2 :

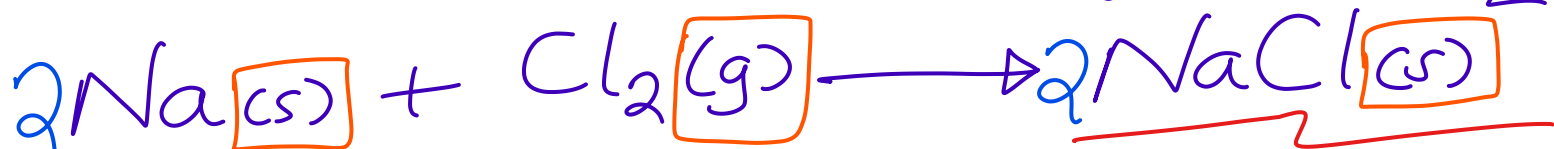


Excess O_2



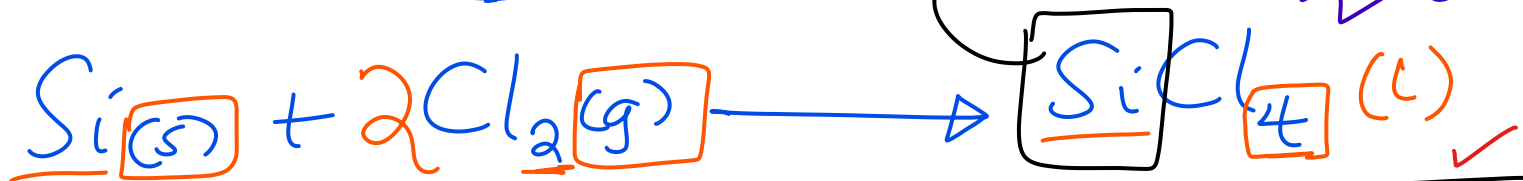
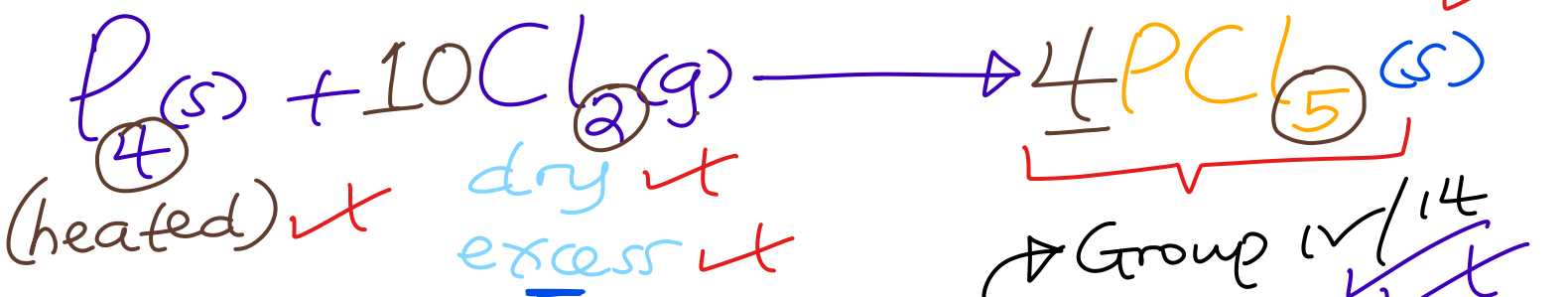
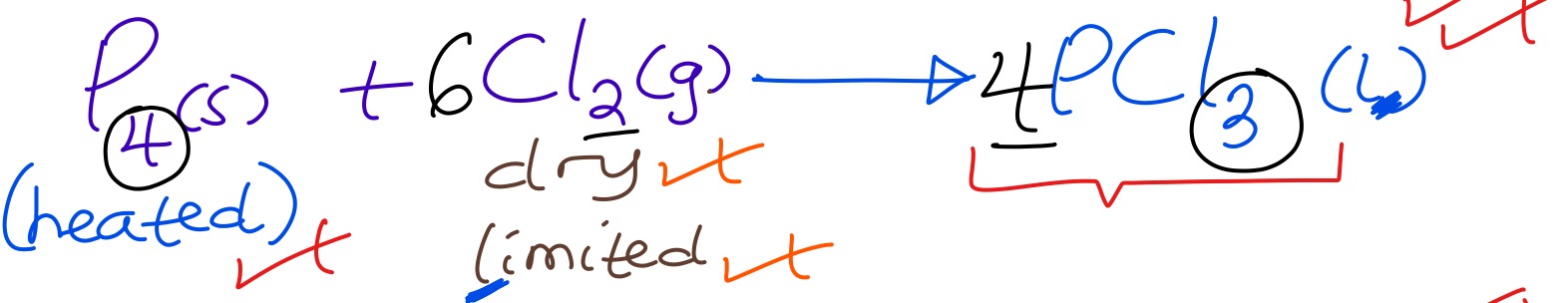
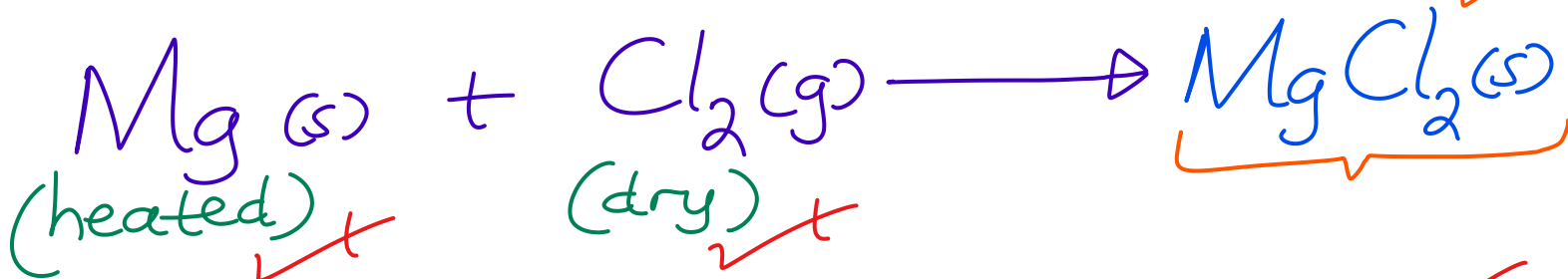
SCIENCE SCHOLARS ACADEMY

* Elements + $\text{Cl}_2 \longrightarrow$ Preparation of chlorides.



heated (hot) sodium ✓
(dry) chlorine ✓

Conditions of reactions!



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Note: Group IV(14) tetrachlorides are LIQUIDS

LASTLY → ✓

SCIENCE SCHOLARS ACADEMY

task: Colour? ✓



Next stop

Reactions with H_2O ✓

connect to those of

dilute Acids *

dilute Alkalis *

Elements. ✓
oxides. ✓
chlorides. ✓

Imresh
Paul Tutor

Science Scholars Academy
UG.

0753279592 {WhatsApp Calls}

ASSIGNMENT: Practice ONLY THE TAUGHT EQNS!

→ End

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